YEAR 7–8 MATHEMATICS

Angles and Measurement

Practice Questions

Work in the space under each question. Give a reason for every angle fact. Answers are in the separate Answers booklet.

SECTION	CORE QUESTIONS ★	ALL	CORE
Part A Angles			
A1 Points, lines, angles and notation	1, 3, 4, 5	11 min	9 min
A2 Measuring and classifying angles	1, 4, 5, 6	10 min	8 min
A3 Angles at a point	1, 2, 3, 5	16 min	11 min
A4 Transversals and parallel lines	1, 2, 4, 5	15 min	12 min
A5 Further parallel-line problems and proving lines parallel	1, 2, 3, 4	16 min	13 min
A6 Triangles	1, 2, 3, 4	16 min	13 min
A7 Quadrilaterals	1, 3, 4	17 min	11 min
Part B Measurement			
B1 Units and conversions	1, 2, 5, 6	13 min	10 min
B2 Perimeter	1, 3, 4	10 min	6 min
B3 Area of rectangles and composite shapes	1, 3, 4	14 min	10 min
B4 Areas of triangles and parallelograms	1, 2, 5, 6	15 min	9 min
B5 Special quadrilaterals: rhombus, kite, trapezium	1, 3, 5	15 min	8 min
Part C Mixed challenge			
C Review and challenge questions	2	29 min	3 min
Total		3 h 17 min	2 h 3 min

This pack holds more than one lesson's worth of questions, so choose questions to suit the student. The questions marked ★ make a suggested core set for a 2-hour lesson: one warm-up in each section, then harder questions. Each section runs from easier to harder. Every question is taken word for word from ICE-EM Mathematics 7 (3rd edition) or Cambridge Maths NSW Year 8 (2nd edition), and its source is shown beside it. Times are a guide for a typical Year 7 or Year 8 student.

PART A

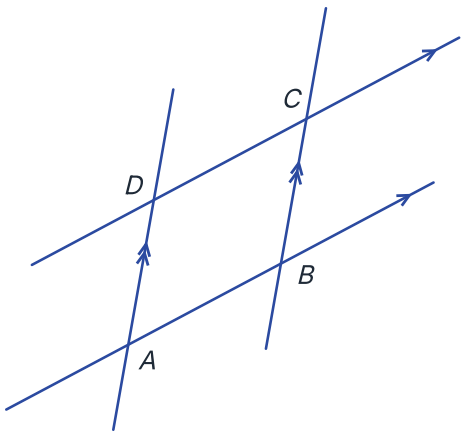
Angles

Points, lines and angle notation; measuring and classifying angles; angles at a point; transversals and parallel lines; proving lines parallel; triangles; quadrilaterals.

A1 POINTS, LINES, ANGLES AND NOTATION · Y7 ICE-EM 6A–6B · Y8 Cambridge 6A

About 11 minutes. Core: questions 1, 3, 4 and 5.

1 ★ The diagram shows two pairs of parallel lines. Y7 ICE-EM Ex 6A Q2



a In symbols, write 'AB is parallel to DC'.

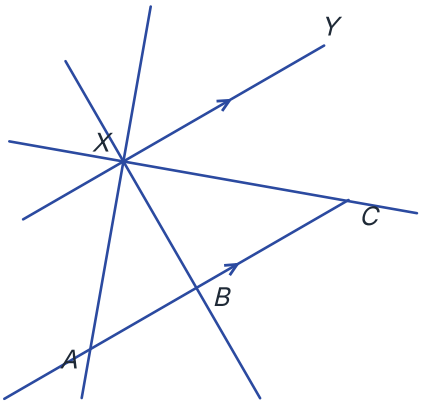
b In symbols, write 'BC is parallel to AD'.

c What is the intersection of the lines AB and BC?

d What point lies on both BC and CD?

e Do any points lie on both lines AB and CD?

2 Look at the diagram. Y7 ICE-EM Ex 6A Q3



a Name the four concurrent lines passing through the point X.

b Name the two parallel lines, using the correct symbol for 'is parallel to'.

c Name the intersection of the lines BX and AC.

d Name the line joining C and the intersection of AX and BX.

e What points belong to both lines AC and CB?

f What points belong to both lines AC and XY?

3 ★ There are five lines in the diagram.

Y7 ICE-EM Ex 6B Q2

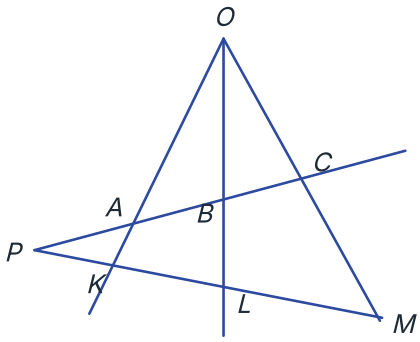
a Copy and complete: The lines AK, BL and CM are _____.

b Does L lie on the ray OB?

c Does L lie on the interval PK?

d Name the two rays with vertex P.

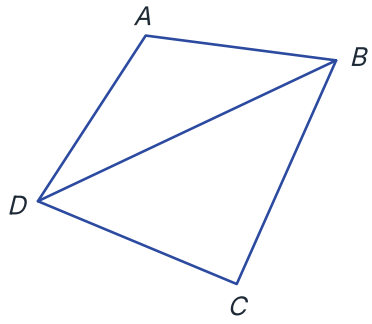
e Name the four rays with vertex A.



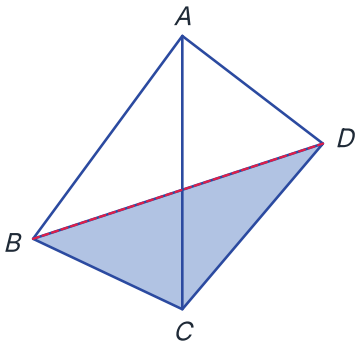
4 ★

Y7 ICE-EM Ex 6B Q7

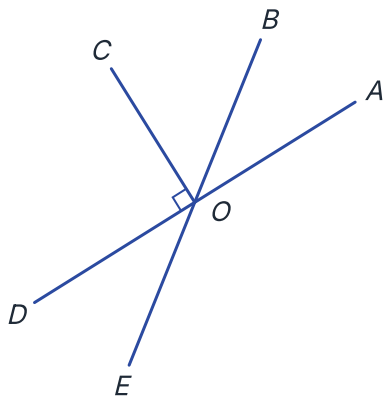
a Name all eight angles associated with the figure below.



b Name all 12 angles formed at the vertices of the tetrahedron below.



5 ★ Consider the given diagram. Y8 Cambridge Ex 6A Q5



a Name an angle that is:
i vertically opposite to $\angle AOB$
ii complementary to $\angle BOC$
iii supplementary to $\angle AOE$
iv supplementary to $\angle AOC$

b Copy and complete:
i $AD \perp$ ___
ii ___ $\perp OD$

A2 MEASURING AND CLASSIFYING ANGLES · Y7 ICE-EM 6C

About 10 minutes. Core: questions 1, 4, 5 and 6.

1 ★ Write down the complement of: Y7 ICE-EM Ex 6C Q1

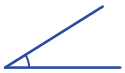
a 15° **b** 35° **c** 72° **d** 88° **e** 56°

2 Write down the supplement of: Y7 ICE-EM Ex 6C Q2

a 160° **b** 100° **c** 75° **d** 88° **e** 146°

3 Classify each angle as acute, obtuse, reflex or right. Y7 ICE-EM Ex 6C Q3

a



b



c



d



4 ★ What is the size of the angle between the hour hand and the minute hand at: Y7 ICE-EM Ex 6C Q4

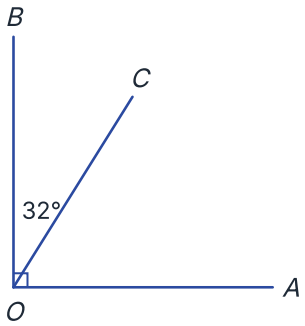
a 6 a.m.? **b** 3 a.m.? **c** 1 p.m.? **d** 2 p.m.?
e 4 a.m.? **f** 8 p.m.? **g** 10 a.m.? **h** 7 a.m.?

5 ★ Through how many degrees does the minute hand of a clock move in: Y7 ICE-EM Ex 6C Q5

a 1 hour? **b** 30 minutes? **c** 45 minutes? **d** 5 minutes?

e 50 minutes? **f** 25 minutes? **g** 2 hours? **h** $1\frac{1}{2}$ hours?

6 ★ Find $\angle AOC$ in the diagram. Also give the size of reflex angle $\angle AOC$. Y7 Ch 6 Review Q4

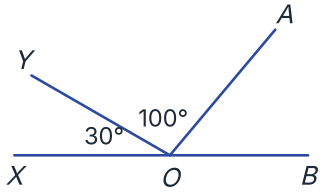


A3 ANGLES AT A POINT · Y7 ICE-EM 6D · Y8 Cambridge 6A

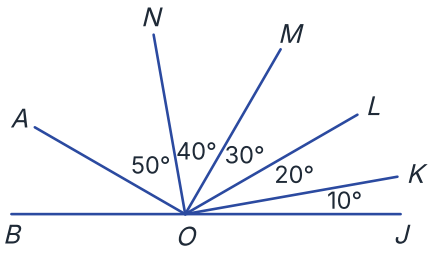
About 16 minutes. Core: questions 1, 2, 3 and 5.

1 ★ Find $\angle AOB$ in each diagram below, giving reasons for your answer. Y7 ICE-EM Ex 6D Q5

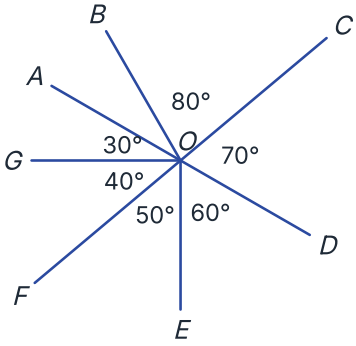
e



k

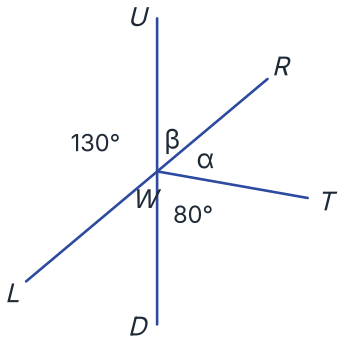


l

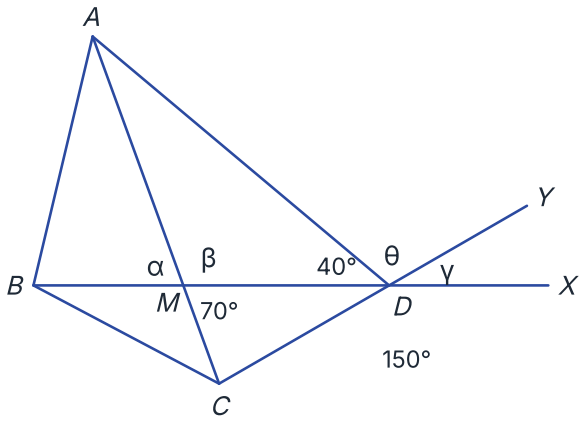


2 ★ Find the values of α, β, γ and θ in each diagram below, carefully giving reasons for all of your statements. Y7 ICE-EM Ex 6D Q7

g

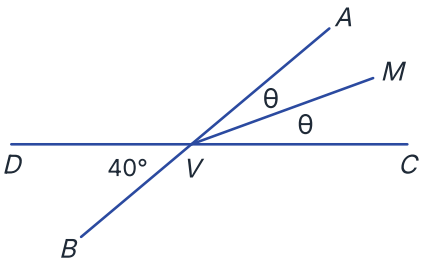


h

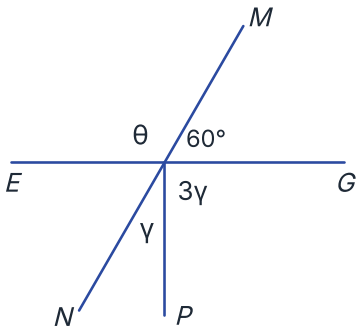


3 ★ In each diagram below, angles marked with the same Greek letter are equal in size. Find the value of each, giving reasons for your answers. Y7 ICE-EM Ex 6D Q8

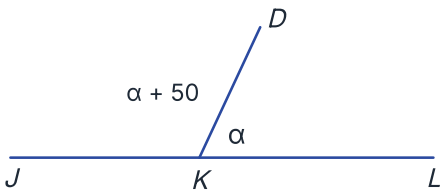
c



g



h



4 A round birthday cake is cut into sectors for nine friends (including Jack) at Jack's birthday party. After the cake is cut, there is no cake remaining. What will be the angle at the centre of the cake for Jack's piece if: Y8 Cambridge Ex 6A Q8

a everyone receives an equal share?

b Jack receives twice as much as everyone else? (In parts **b**, **c** and **d** assume his friends have an equal share of the rest.)

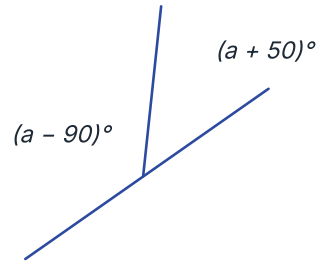
c Jack receives four times as much as everyone else?

d Jack receives ten times as much as everyone else?

5 ★ Consider this diagram (not drawn to scale). Y8 Cambridge Ex 6A Q13

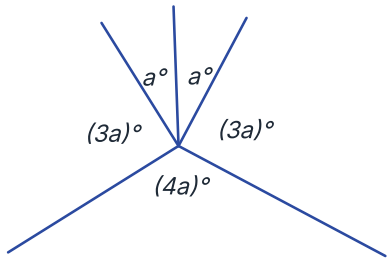
a Calculate the value of a.

b Explain what is wrong with the way the diagram is drawn.

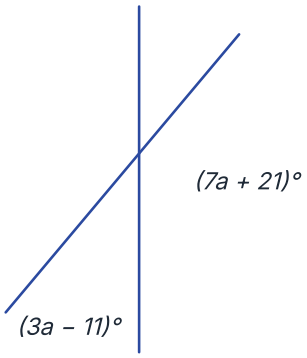


6 Equations can be helpful in solving geometric problems in which more complex expressions are involved. Find the value of a in these diagrams. Y8 Cambridge Ex 6A Q14

e



f



A4 TRANSVERSALS AND PARALLEL LINES · Y7 ICE-EM 6E · Y8 Cambridge 6B

About 15 minutes. Core: questions 1, 2, 4 and 5.

1 ★ Name the angle that is: Y8 Cambridge Ex 6B Q2

a corresponding to $\angle ABF$

b corresponding to $\angle BCG$

c alternate to $\angle FBC$

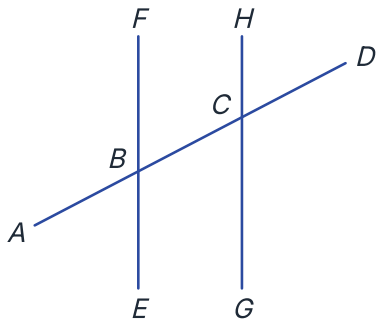
d alternate to $\angle CBE$

e cointerior to $\angle HCB$

f cointerior to $\angle EBC$

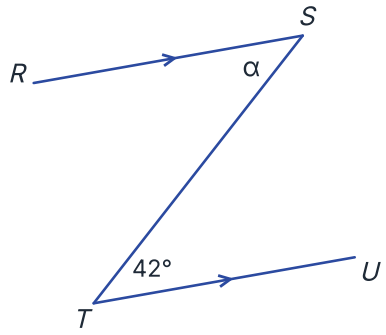
g vertically opposite to $\angle ABE$

h vertically opposite to $\angle HCB$

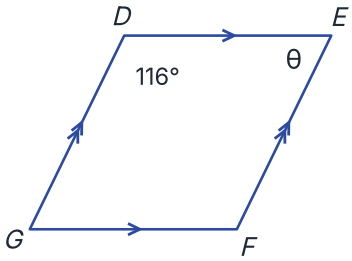


2 ★ Find the values of α, β, γ and θ in the diagrams below. Give careful reasons for all your statements, mentioning the relevant parallel lines. Y7 ICE-EM Ex 6E Q7

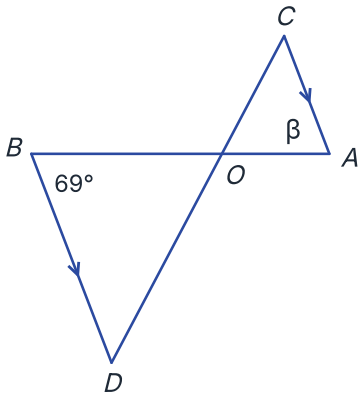
d



g

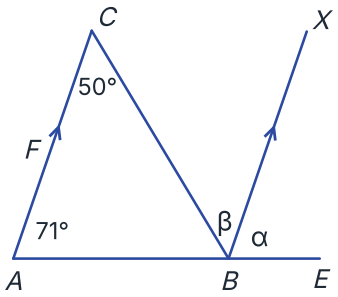


k

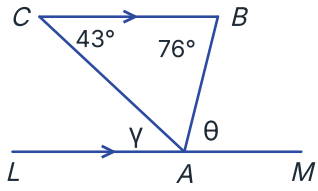


3 Find the values of the letters α, β, γ and θ in the diagrams below. Give a reason for each step in your argument, and name the relevant parallel lines. Y7 ICE-EM Ex 6E Q8

f

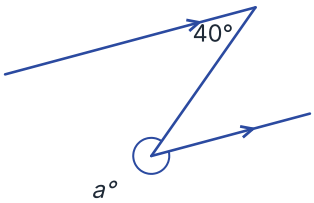


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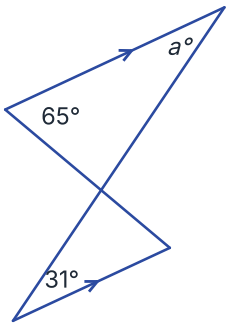


4 ★ Find the value of a in these diagrams. Y8 Cambridge Ex 6B Q7

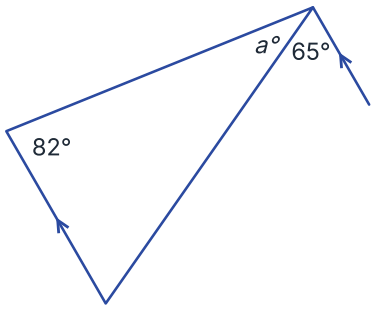
b



e

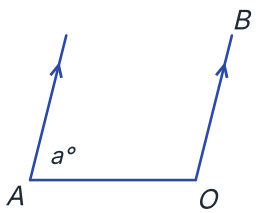


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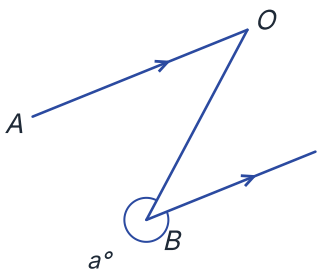


5 ★ Write an algebraic expression for the size of $\angle AOB$. Y8 Cambridge Ex 6B Q9

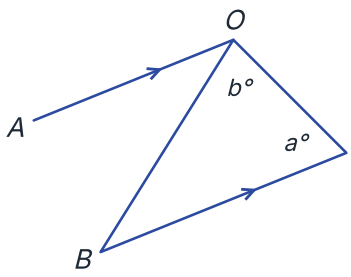
a



b



c

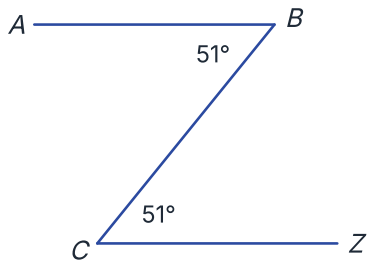


A5 FURTHER PROBLEMS AND PROVING LINES PARALLEL · Y7 ICE-EM 6F–6G

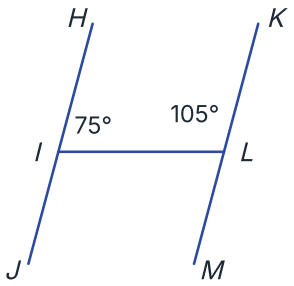
About 16 minutes. Core: questions 1, 2, 3 and 4.

1 ★ For each diagram below, name all pairs of parallel lines, giving reasons. Y7 ICE-EM Ex 6G Q1

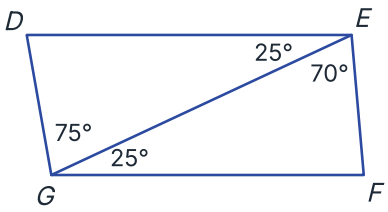
a



c

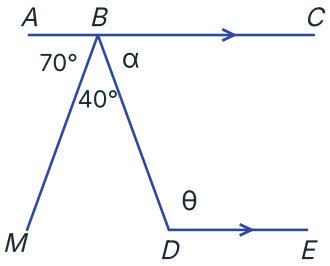


f

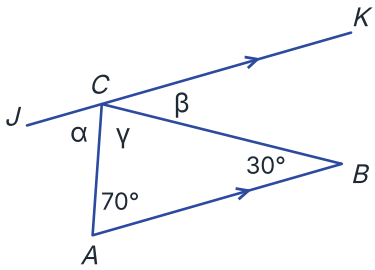


2 ★ Find the values of α, β, γ and θ in each of the diagrams below. Give careful reasons for all your statements. Y7 ICE-EM Ex 6F Q1

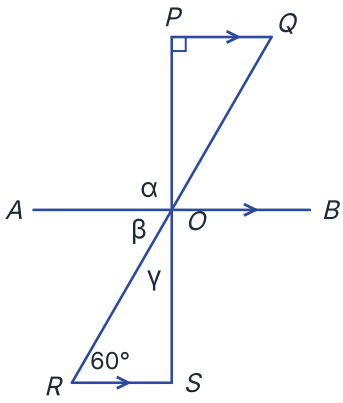
c



e



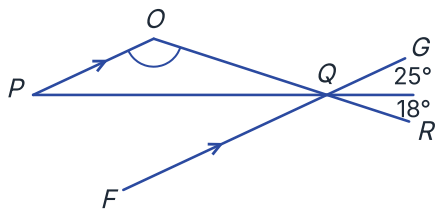
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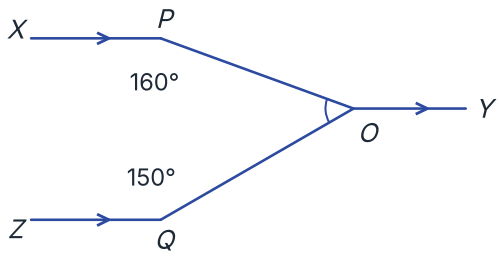
3 ★ Find the size of the marked angle $\angle POQ$ in each diagram below. Each solution will require two or more steps, each with its own reason.

Y7 ICE-EM Ex 6F Q2

b



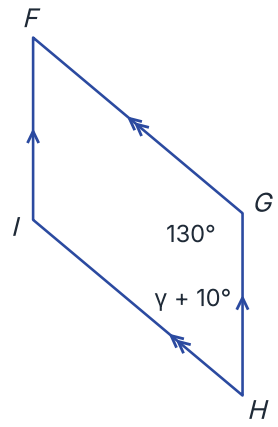
d



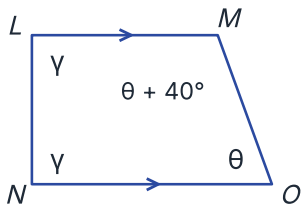
4 ★ Find the values of α, β, γ and θ, giving reasons for your answers.

Y7 ICE-EM Ex 6F Q4

d

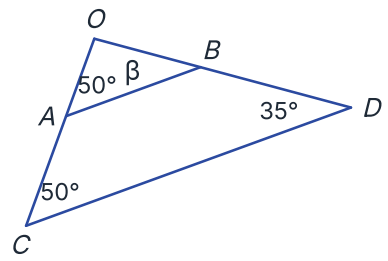


h

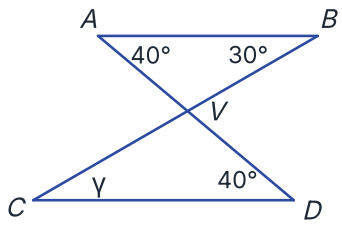


5 In each diagram below, give a reason why $AB \parallel CD$. Hence, find the values of α, β, γ and θ. Give all reasons. Y7 ICE-EM Ex 6G Q2

b



d



A6 TRIANGLES · Y8 Cambridge 6C

About 16 minutes. Core: questions 1, 2, 3 and 4.

1 ★ Give the common name of a triangle with these properties. Y8 Cambridge Ex 6C Q1

a one right angle

b two equal side lengths

c all angles acute

d all angles 60°

e one obtuse angle

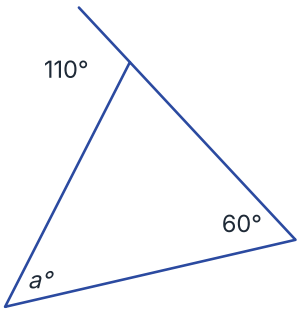
f three equal side lengths

g two equal angles

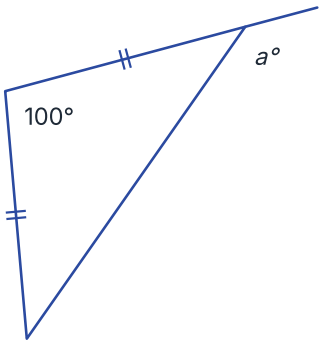
h three different side lengths

2 ★ Find the value of a. Y8 Cambridge Ex 6C Q7

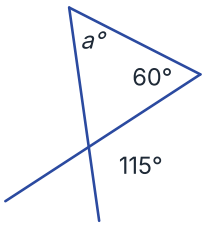
d



e

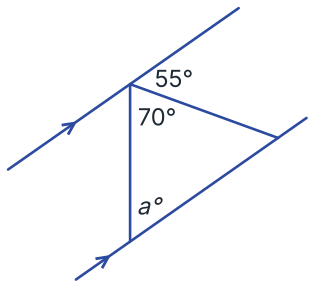


f

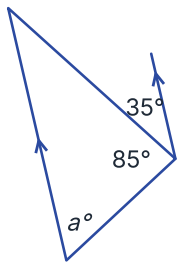


3 ★ Use your knowledge of parallel lines and triangles to find the unknown angle a. Y8 Cambridge Ex 6C Q9

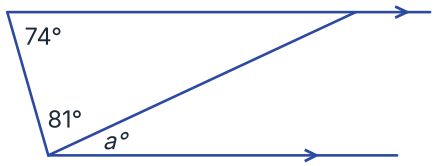
a



b

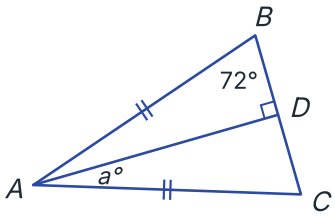


c

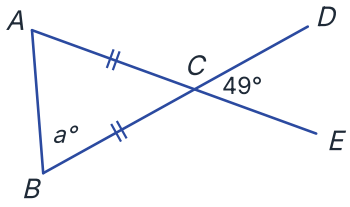


4 ★ Find the value of a in these diagrams, giving reasons. Y8 Cambridge Ex 6C Q10

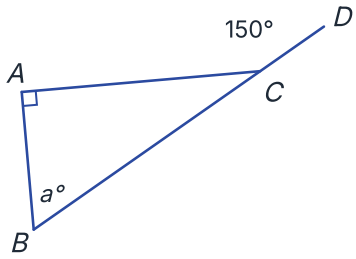
c



e



f



5 A triangle is constructed using two radii and a chord. Y8 Cambridge Ex 6C Q11

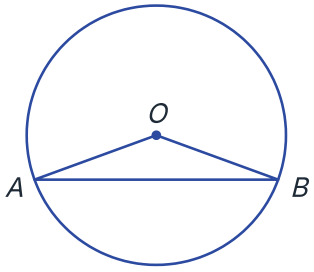
a What type of triangle is $\triangle AOB$? Explain.

b Name two angles that are equal.

c Find $\angle ABO$ if $\angle BAO$ is 30°.

d Find $\angle AOB$ if $\angle OAB$ is 36°.

e Find $\angle ABO$ if $\angle AOB$ is 100°.

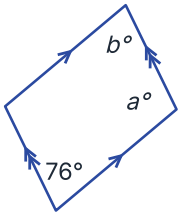


A7 QUADRILATERALS · Y8 Cambridge 6D

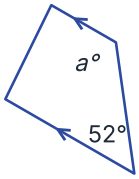
About 17 minutes. Core: questions 1, 3 and 4.

1 ★ Find the value of the pronumerals in these quadrilaterals. Y8 Cambridge Ex 6D Q3

a

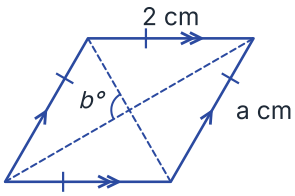


c

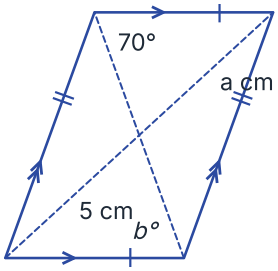


2 By considering the properties of the given quadrilaterals, give the values of the pronumerals. Y8 Cambridge Ex 6D Q5

b

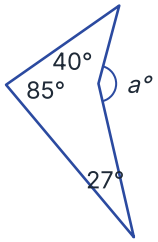


c

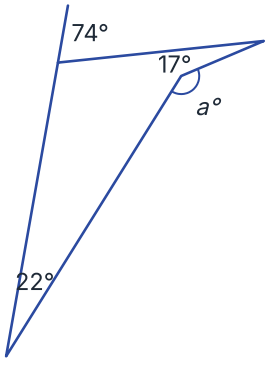


3 ★ Use your knowledge of geometry from the previous sections to find the values of a. Y8 Cambridge Ex 6D Q6

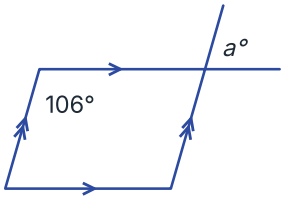
a



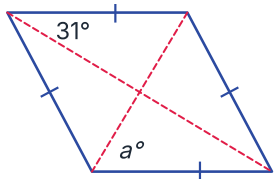
c



d

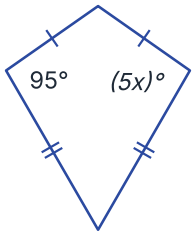


e

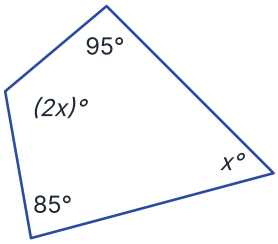


4 ★ Use your algebra skills to find the value of x in these diagrams. Y8 Cambridge Ex 6D Q7

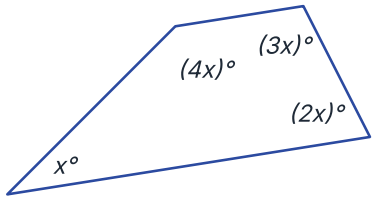
a



b



c



5 By considering the properties of special quadrilaterals, decide if the following are always true. Y8 Cambridge Ex 6D Q9

a A square is a type of rectangle.

b A rectangle is a type of square.

c A square is a type of rhombus.

d A rectangle is a type of parallelogram.

e A parallelogram is a type of square.

f A rhombus is a type of parallelogram.

6 Is it possible to draw a non-convex quadrilateral with two or more interior reflex angles? Explain and illustrate. Y8 Cambridge Ex 6D Q10

PART B

Measurement

Units and conversions; perimeter; areas of rectangles and composite shapes; areas of triangles and parallelograms; areas of the rhombus, kite and trapezium.

B1 UNITS AND CONVERSIONS · Y7 ICE-EM 9A–9B

About 13 minutes. Core: questions 1, 2, 5 and 6.

1 ★ Copy and complete these statements. Y7 ICE-EM Ex 9A Q2

e 27 mm = ____ cm **h** 1.9 m = ____ cm **k** 6.34 km = ____ m

l 200 000 mm = ____ km **n** 2901 m = ___ km **s** $\frac{5}{8}$ cm = ____ mm

2 ★ Markers on the road are placed 20 m apart. If the distance from the first marker to the last one is 1 km, how many markers are there in total? Y7 ICE-EM Ex 9A Q9

3 A picture, which is 60 cm wide, is to be placed in the centre of a wall. The length of the wall from left to right is 5 m. What is the distance between the left-hand edge of the picture and the left-hand edge of the wall? Y7 ICE-EM Ex 9A Q10

4 Peter walks 5.1 km each day. How many micrometres is this? Y7 ICE-EM Ex 9B Q4

5 ★ How many minutes are there in a micro-century? (Assume there are 36 525 days in a century.) Y7 ICE-EM Ex 9B Q8

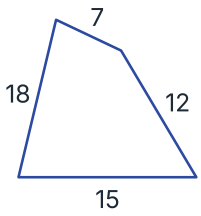
6 ★ In bygone days, the imperial system of measurement was used. In this system, distance is measured in inches, feet, yards and miles. There are 12 inches in a foot, 3 feet in a yard, and 1760 yards in a mile. Given that 2.54 cm equals 1 inch, how many kilometres is a mile? Y7 Ch 9 Review Q22

B2 PERIMETER · Y7 ICE-EM 9D

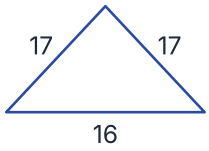
About 10 minutes. Core: questions 1, 3 and 4.

1 ★ Find the perimeter of each of these figures. Sides that are equal are marked with the same symbol. Y7 ICE-EM Ex 9D Q1

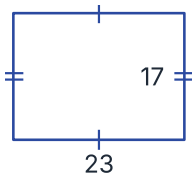
d



e



f



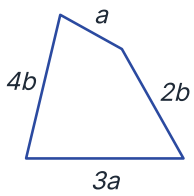
2 Calculate the perimeter of a rectangular room with dimensions of: Y7 ICE-EM Ex 9D Q4

a length 2.3 m and width 3.6 m

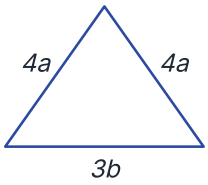
b length 9.8 m and width 6.4 m

3 ★ Write an algebraic expression for the perimeter of each figure. Y7 ICE-EM Ex 9D Q5

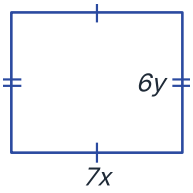
d



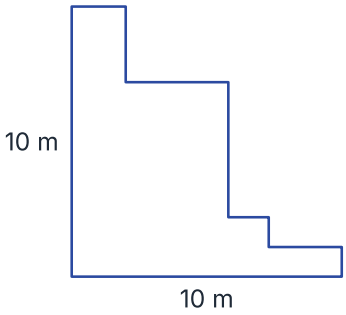
e



f



4 ★ Find the perimeter of the figure below. All angles are right angles. Y7 ICE-EM Ex 9D Q8



5 A fence is required for a rectangular block of land that measures 51.6 m down each of the two sides and 16.85 m across the back, with no fence at the front. How long is the fence in total? Y7 Ch 9 Review Q5

6 What is the perimeter, in metres, of a regular pentagon (all sides are equal), with side length 1.3 mm? Y7 Ch 9 Review Q14

B3 AREA OF RECTANGLES AND COMPOSITE SHAPES · Y7 ICE-EM 9E–9F

About 14 minutes. Core: questions 1, 3 and 4.

1 ★ Y7 ICE-EM Ex 9E Q4

a Find the area in square metres of a rectangle with length 5 m and width 85 cm.

b Find the area in square kilometres of a rectangular region of land with length 800 m and width 2 km.

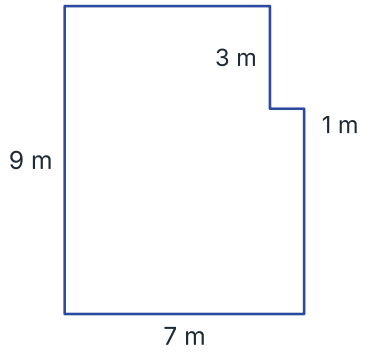
2 The table below gives incomplete information for four rectangles. Copy the table and fill in the gaps. Y7 ICE-EM Ex 9E Q7

	Length	Width	Area
a	7 cm	6 cm	
b	9 cm		27 cm^2
c		17 m	85 m^2
d	12 km		144 km^2

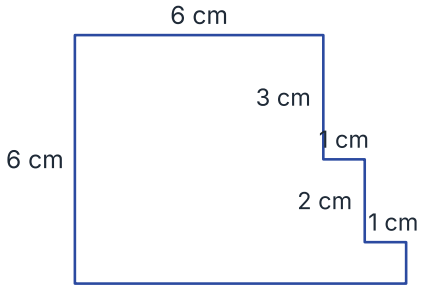
3 ★ The floors of three rooms of a palace are to be covered in tiles. Each tile is one centimetre square and costs $4.50. The rooms measure 8 m by 5.4 m, 2.7 m by 8.9 m and 300 cm by 13.3 m, respectively. Calculate the total cost. Y7 ICE-EM Ex 9E Q15

4 ★ Calculate the area of each figure. Y7 ICE-EM Ex 9F Q2

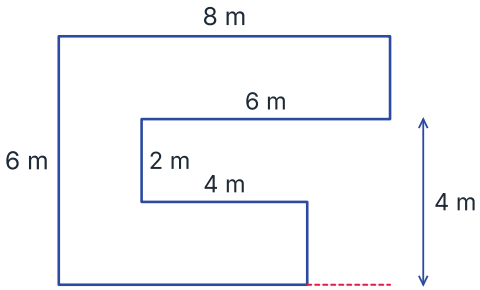
d



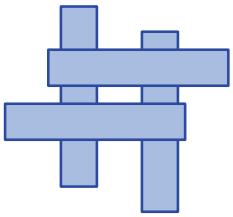
f



h



5 Four rectangles, each with length 5 m and width 1 m, overlap to form the figure shown. Calculate its area. Y7 ICE-EM Ex 9F Q5

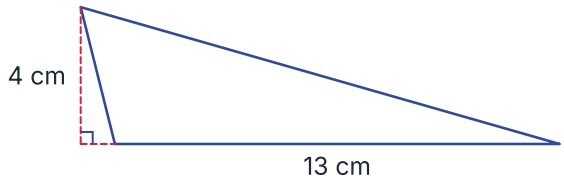


B4 AREAS OF TRIANGLES AND PARALLELOGRAMS · Y7 ICE-EM 9G

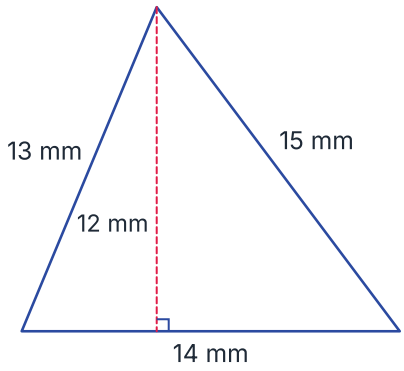
About 15 minutes. Core: questions 1, 2, 5 and 6.

1 ★ Find the area of each triangle. Y7 ICE-EM Ex 9G Q1

c

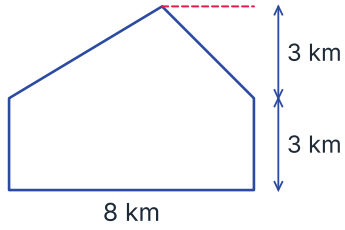


d

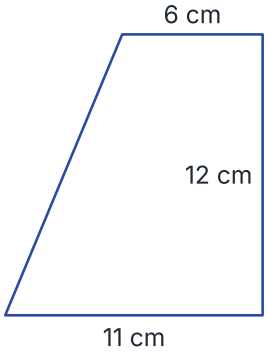


2 ★ Find the area of each figure below by adding the areas of squares, rectangles and triangles as necessary. Y7 ICE-EM Ex 9G Q7

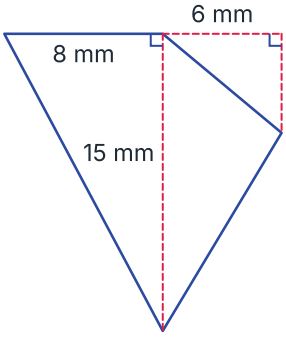
b



c

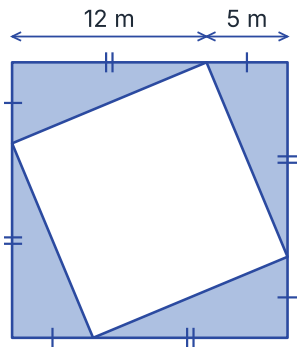


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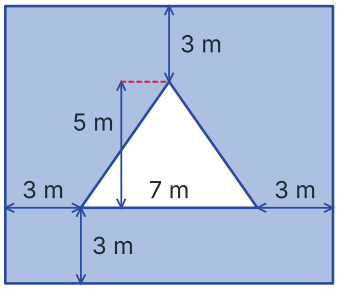


3 Find the area of the shaded region in each figure below by adding and subtracting the areas of squares, rectangles and triangles as necessary. Y7 ICE-EM Ex 9G Q8

a

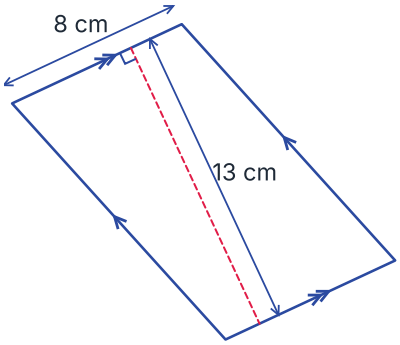


d

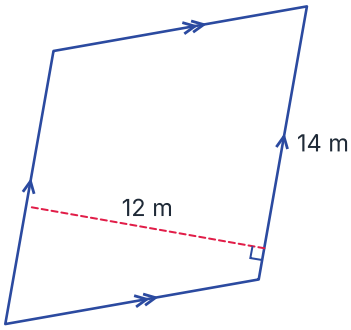


4 Find the areas of these regions. Y7 ICE-EM Ex 9G Q11

a

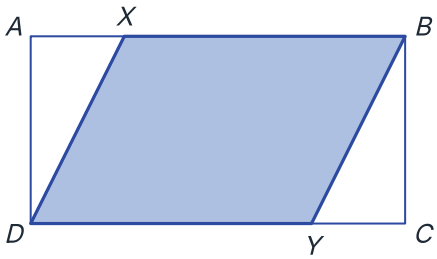


b



5 ★ A parallelogram has a height that is half of its base. The area of the parallelogram is 18 cm^2. What is the height and base of the parallelogram? Y7 ICE-EM Ex 9G Q14

6 ★ Rectangle $ABCD$ has length 8 cm and height 4 cm. Y7 ICE-EM Ex 9G Q15



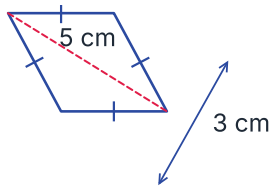
The two triangles AXD and BYC have height 4 cm and base 2 cm. Find the area of the parallelogram $DXBY$.

B5 SPECIAL QUADRILATERALS · Y8 Cambridge 3D

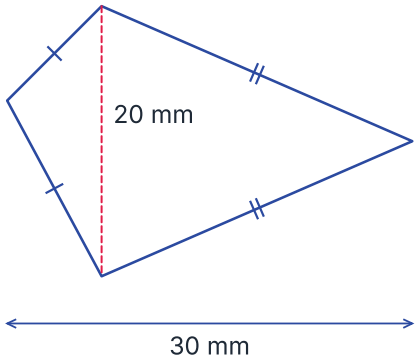
About 15 minutes. Core: questions 1, 3 and 5.

1 ★ Find the area of these special quadrilaterals. First give the name of the shape. Y8 Cambridge Ex 3D Q3

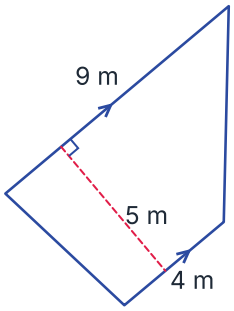
d



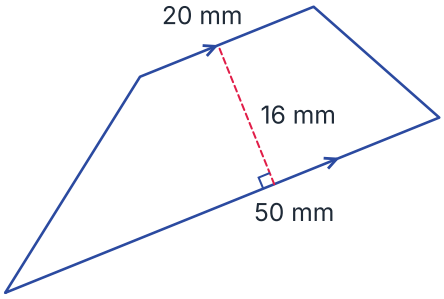
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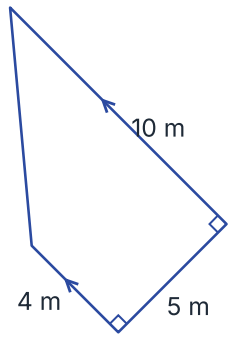


l

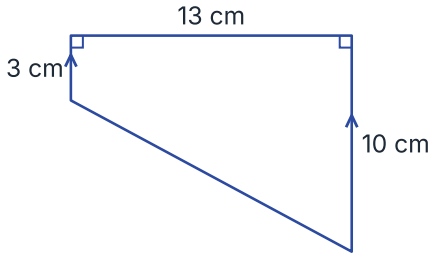


2 These trapeziums have one side at right angles to the two parallel sides. Find the area of each. Y8 Cambridge Ex 3D Q4

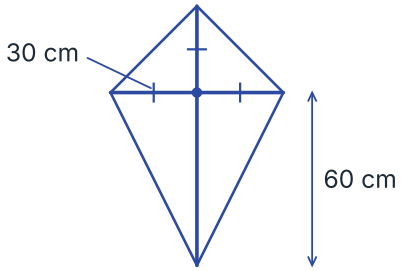
b



c



3 ★ A flying kite is made from four centre rods all connected near the middle of the kite, as shown. What area of plastic, in square metres, is needed to cover the kite? Y8 Cambridge Ex 3D Q5

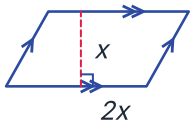


4 A landscape gardener charges $\$20$ per square metre of lawn. A lawn area is in the shape of a rhombus and its diagonals are 8 m and 14.5 m. What would be the cost of laying this lawn? Y8 Cambridge Ex 3D Q7

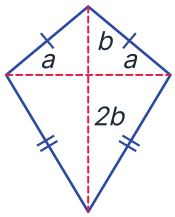
5 ★ The parallel sides of a trapezium are 2 cm apart and one of the sides is three times the length of the other. If the area of the trapezium is 12 cm^2, what are the lengths of the parallel sides? Y8 Cambridge Ex 3D Q8

6 Write an expression for the area of these shapes in simplest form (e.g. $A = 2a + 3ab$). Y8 Cambridge Ex 3D Q10

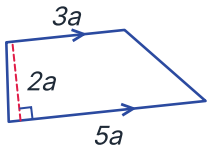
a



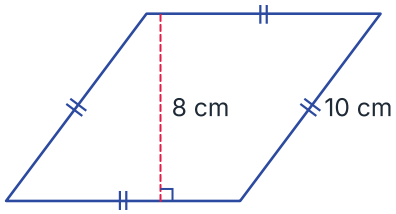
b



c



7 Would you use the formula $A = \frac{1}{2}xy$ to find the area of this rhombus? Explain. Y8 Cambridge Ex 3D Q11



PART C

Mixed challenge

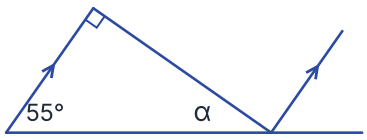
Review and challenge questions from the two Year 7 chapters: angles, then length, perimeter and area.

C MIXED CHALLENGE · Y7 ICE-EM Ch 6 and Ch 9 Review and Challenge

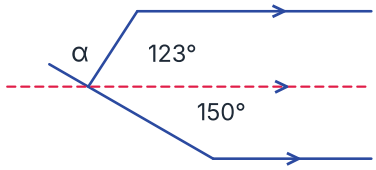
About 29 minutes. Core: question 2.

1 Find the value of α in each diagram below. Y7 Ch 6 Review Q7

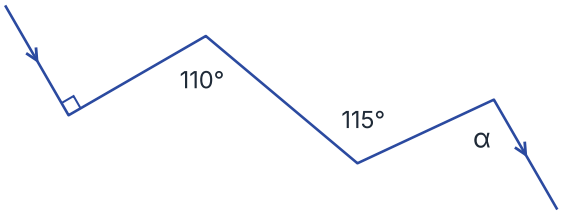
b



d

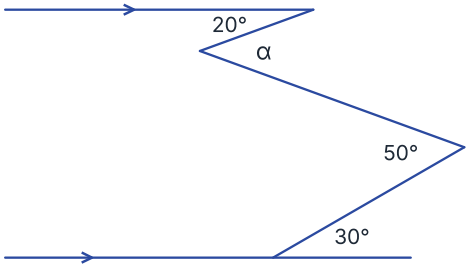


2 ★ Calculate the unknown angle. Y7 Ch 6 Challenge Q1

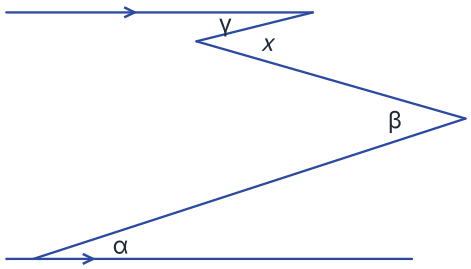


3 Y7 Ch 6 Challenge Q2

a Find the value of α.



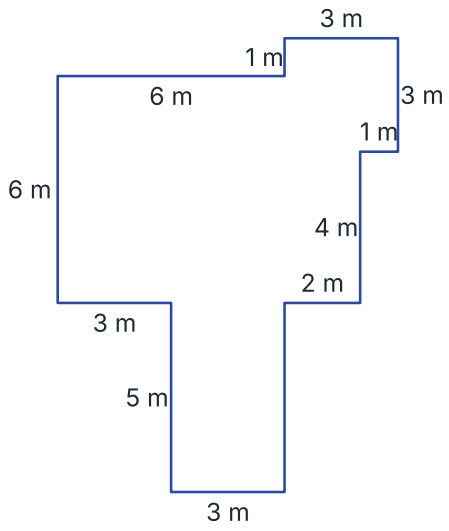
b Find x in terms of α, β and γ.



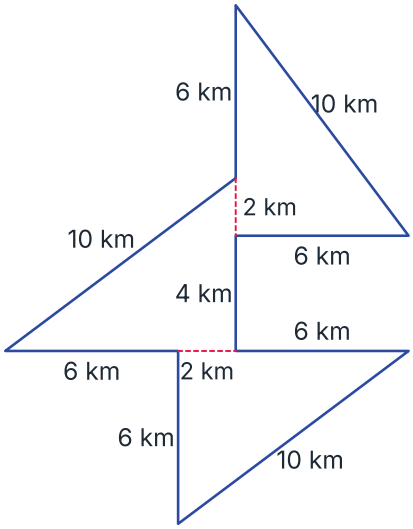
4 How many minutes pass after 12 noon before the angle between the minute and hour hand is exactly 90°? Y7 Ch 6 Challenge Q7

5 Calculate the perimeter and area of each of these figures, assuming that the non-slanting lines are either horizontal or vertical. Y7 Ch 9 Review Q8

a



b



6 Triangle ABC has side lengths $AC = 4$ cm, $AB = 3$ cm and $BC = 5$ cm. $\angle BAC = 90°$. The point X is on BC and AX is perpendicular to BC. Find: Y7 Ch 9 Review Q26

a the area of triangle ABC

b the length AX

7 A path 1 metre wide is being placed around the grassed area of a garden, as shown below. What is the area of the grassed part of the garden? What is its perimeter? What will the area of the path be?

Y7 Ch 9 Challenge Q4

